CAMI's Theorems and Theories

Volume III.2: *Iterated Location Reports and Higher-Order Boards*

A mathematical companion beginning with Volume III.1

CAMI (AI-assisted mathematical exposition)
Research interlocutor: Brian Tenneson

October 15, 2026
PRELIMINARY RESEARCH NOTE

Abstract

Starting with a finite board A_0 and a fixed alphabet $B = P \cup \{E\}$, we study the exact recursive construction $A_{n+1} = [A_n \to B]$. An element of A_1 is a complete occupancy report on the original board; at every subsequent stage, an entire report becomes one of the index locations of the next board. We identify the resulting higher-order function-space objects, derive their cardinal growth, and prove an iterated-power-set correspondence for two tokens. We distinguish this *board-of-reports tower* from a different and equally meaningful interpretation in which every original square can contain a whole previous-level report. Finally we show that the recursion alone provides increased expressive capacity, not certified physical or epistemic resolution; coherent refinement requires explicit decoding maps. All theorems are elementary, conditional where needed, and make no historical-priority claim.

"A report can become a square without becoming a piece."

Version and terminology. This is a new companion to the corrected Volume III.1, not a replacement or reordering of its nine sections. A single location report is a *function*; the collection of all such functions is a *set of reports*. The empty-square token is always E, never a set-theoretic empty object.

Contents

1 The question and its exact recursion **2**

2 What kind of objects are the A_n? 2

3 The cardinal-growth theorem 3

4 Binary location reports and the iterated power set 3

5 A second meaning of "every square contains a report" 4

6 When is the tower a resolution enhancement? 4

7 Relevance to Volume III.1 and certified knowledge 5

8 Conclusions, provenance, and next theorem target 6

1 The question and its exact recursion

Let A_0 be a given finite set of squares (possibly empty), let P be a set of occupant tokens, and let $E \notin P$ denote the empty-square token. Put $B = P \cup \{E\}$; this is the alphabet denoted P' in the question. For sets X, Y, $[X \to Y]$ means the set of *all* total functions from X to Y.

Definition 1.1 (Board-of-reports tower)**.** Define recursively

$$\boxed{A_{n+1} := [A_n \to B], \qquad n \geq 0.}$$

An element $x \in A_n$ has *type* n. An element $F \in A_{n+1}$ is called a level-$(n+1)$ report and satisfies $F : A_n \to B$.

In particular, $A_1 = [A_0 \to B]$ is the collection of *all* complete first-level reports, and $L \in A_1$ is just one such report. The next-level report $F \in A_2$ assigns one of the original labels in B to *each first-level report* $L \in A_1$. Those earlier reports are now the domain or "squares" of the next board, *not* the allowable occupant labels.

Proposition 1.2 (Higher-order typing)**.** *For each $n \geq 1$, an element $F \in A_n$ is a total map $F : A_{n-1} \to B$. It assigns B-valued labels to all objects of type $n-1$. Thus $A_2 = [[A_0 \to B] \to B]$ and $A_3 = [[[A_0 \to B] \to B] \to B]$, with the indicated parenthesization.*

Proof. Immediate by induction from Definition 1.1. □

Remark 1.3 (All functions versus legal positions)**.** The displayed tower is *unrestricted*: it includes functions that might violate chess occupancy constraints (such as repeated individual identities). Imposing admissibility predicates produces selected subsets and, if those subsets become subsequent domains, a *different* recursion. The theorem statements here concern the unrestricted spaces unless expressly qualified.

2 What kind of objects are the A_n?

Theorem 2.1 (Function spaces, not physical boards)**.** *Each A_n is a set. For $n \geq 1$, it is a set of higher-order, B-valued functions on the immediately preceding set A_{n-1}. The construction needs no legal-move relation, chronology, probability, verifier, or distinguished actual location report. In particular, $F \in A_2$ by itself asserts nothing about a move between the reports that belong to A_1.*

Proof. Construction of the function space is set-theoretic. No transition or event relation occurs in the recursion, so it cannot imply the existence or occurrence of one. □

Remark 2.2 (Types and overlap)**.** A statement $x \in A_n$ identifies a *level*, even if underlying set encodings accidentally have overlaps. For an explicitly disjoint typed universe, use $\bigsqcup_{n \geq 0}(\{n\} \times A_n)$. Neither $A_n \subseteq A_{n+1}$ nor a canonical projection $A_{n+1} \to A_n$ is part of Definition 1.1.

Example 2.3 (A board with two squares and three occupant labels)**.** Let $A_0 = \{a, b\}$ and $B = \{E, p, q\}$. Then A_1 consists of all $3^2 = 9$ functions $\{a, b\} \to B$. A particular $F \in A_2$ gives a label E, p, or q to each of these nine *entire* reports. Therefore $|A_2| = 3^9 = 19{,}683$. This does *not* mean that each original square now holds one of 19,683 pieces.

3 The cardinal-growth theorem

Theorem 3.1 (Exponentiation at every level)**.** *Suppose A_0 and B are finite, $|A_0| = m$, and $|B| = k \geq 1$. Put $a_n = |A_n|$. Then*

$$\boxed{a_0 = m, \qquad a_{n+1} = k^{a_n}.}$$

In particular, $a_1 = k^m$, $a_2 = k^{k^m}$, and $a_3 = k^{k^{k^m}}$, where powers associate from the top down.

Proof. For finite X of size a, there are exactly k^a functions $X \to B$. Apply this at each level. □

Theorem 3.2 (Strict growth when there is at least one genuine piece)**.** *If $P \neq \varnothing$, then $|A_{n+1}| > |A_n|$ for every $n \geq 0$, even if the original alphabet or a board-level set is infinite. Thus the tower never stabilizes in cardinality.*

Proof. Choose $p \in P$, so the two labels E, p are distinct. Every subset $S \subseteq A_n$ gives an indicator $\chi_S : A_n \to B$ taking the value p on S and E elsewhere. This injects $\mathcal{P}(A_n)$ into A_{n+1}. Cantor's theorem yields $|\mathcal{P}(A_n)| > |A_n|$; hence $|A_{n+1}| > |A_n|$. □

Remark 3.3 (Degenerate one-token case)**.** If $P = \varnothing$, then $B = \{E\}$ and A_1 is a singleton for every A_0, including $A_0 = \varnothing$. All subsequent A_n are singletons. Thus the nonempty-piece assumption in Theorem 3.2 is essential.

Example 3.4 (A small start becomes enormous)**.** For two original squares and two possible labels ($m = k = 2$), the successive cardinalities are

$$|A_0| = 2, \quad |A_1| = 4, \quad |A_2| = 16, \quad |A_3| = 65{,}536, \quad |A_4| = 2^{65{,}536}.$$

This is growth in the number of *possible functions*, not the creation of physical space or experimentally acquired knowledge.

4 Binary location reports and the iterated power set

Theorem 4.1 (Characteristic-function equivalence)**.** *If $B = \{E, p\}$, then for every $n \geq 0$ there is a canonical bijection, relative to the selected label p,*

$$A_{n+1} \longleftrightarrow \mathcal{P}(A_n), \qquad F \longmapsto F^{-1}(\{p\}).$$

Consequently the tower is levelwise isomorphic to the iterated power-set hierarchy based on A_0, with its levels kept distinct.

Proof. Given $S \subseteq A_n$, define $F_S(x) = p$ if $x \in S$ and $F_S(x) = E$ otherwise. The assignments $F \mapsto F^{-1}(\{p\})$ and $S \mapsto F_S$ are inverse bijections. Iteration proves the assertion. □

Remark 4.2 (More than two labels)**.** For finite B of size $k > 2$, each $F : A_n \to B$ instead describes a labeled partition of A_n into the disjoint fibers $F^{-1}(b)$ ($b \in B$), allowing empty fibers. The union of the fibers is A_n. It is not simply a single subset of A_n.

5 A second meaning of "every square contains a report"

The question suggests that *every square might contain every element of* A_1. If "square" still means a member of the *original* board A_0, this is not the same construction as Definition 1.1.

Definition 5.1 (Nested-occupant tower)**.** Set $C_1 := [A_0 \to B] = A_1$ and, for $n \geq 1$, define

$$\boxed{C_{n+1} := [A_0 \to C_n].}$$

A member $G \in C_2 = [A_0 \to A_1]$ assigns a *whole first-level report* $G(a) \in A_1$ to every *original square* $a \in A_0$.

Theorem 5.2 (The towers encode different questions)**.** *With* $|A_0| = m \geq 1$ *and* $|B| = k \geq 2$ *finite,*

$$|A_{n+1}| = k^{|A_n|}, \qquad |C_n| = k^{m^n} \quad (n \geq 1).$$

The types also differ: $A_2 = [A_1 \to B]$ *uses earlier* reports as domain locations*, whereas* $C_2 = [A_0 \to A_1]$ *uses earlier reports as* occupant values *on the original domain. Equality of their cardinalities in a special example would not identify these different roles.*

Proof. The first formula is Theorem 3.1. For the second, $|C_1| = k^m$ and if $|C_n| = k^{m^n}$ then $|C_{n+1}| = |C_n|^m = k^{m^{n+1}}$. The descriptions of the function domains and codomains follow from the definitions. □

Example 5.3. With two original squares and three labels, $|C_2| = 9^2 = 81$ while $|A_2| = 3^9 = 19{,}683$. In C_2, each of two old squares holds one of nine old reports; in A_2, each of nine old reports is assigned one of three tokens.

Remark 5.4 (Optional empty occupant at the new level)**.** If a square can either hold an entire first-level report or be genuinely empty, replace the codomain A_1 of C_2 by the tagged set $A_1 \sqcup \{E_2\}$, where E_2 is distinguished from every function in A_1. This is an additional modeling decision. Reusing E without typing it risks confusing absence of an embedded report with the E-valued cells *inside* a report.

6 When is the tower a resolution enhancement?

An increase in the number of representable objects is not, by itself, a gain in *information about one fixed world*. Even knowing $F \in A_2$ does not identify a "current" $L \in A_1$. Compare maps only after specifying their meaning.

Definition 6.1 (Coherent lift by a decoding map)**.** Choose a map $\delta_n : A_{n+1} \to A_n$. It defines a lift of previous-level reports by precomposition:

$$\iota_n : A_{n+1} \to A_{n+2}, \qquad \iota_n(f) = f \circ \delta_n.$$

Here $f : A_n \to B$, $\delta_n : A_{n+1} \to A_n$, so $f \circ \delta_n : A_{n+1} \to B$. A chosen δ_n says how a new "square" is interpreted as an old one.

Theorem 6.2 (Exact criterion for faithful lifting)**.** *Assume* $|B| \geq 2$*. The lift* ι_n *in Definition 6.1 is injective if and only if* δ_n *is surjective.*

Proof. If δ_n is onto and $f \circ \delta_n = g \circ \delta_n$, every $a \in A_n$ has a preimage x, giving $f(a) = f(\delta_n(x)) = g(\delta_n(x)) = g(a)$. Conversely, if δ_n is not onto, choose $a \in A_n$ outside its image. There are distinct functions $f, g : A_n \to B$ that differ only at a, since B has two labels; yet $f \circ \delta_n = g \circ \delta_n$. □

Definition 6.3 (Levelwise Hamming distance)**.** If A_n is finite, for $f, g \in A_{n+1}$ define

$$d_n(f,g) := |\{a \in A_n : f(a) \neq g(a)\}|.$$

Theorem 6.4 (Exact distance under a surjective lift)**.** *If A_n, A_{n+1} are finite and $\delta_n : A_{n+1} \to A_n$ is onto, then for $f, g \in A_{n+1}$,*

$$\boxed{d_{n+1}(\iota_n(f), \iota_n(g)) = \sum_{a \in A_n} |\delta_n^{-1}(a)|\, \mathbf{1}_{f(a)\neq g(a)} \;\geq\; d_n(f,g).}$$

If every fiber of δ_n has the same size r, the left-hand side equals $r\, d_n(f,g)$.

Proof. The lifted functions disagree at $x \in A_{n+1}$ exactly if $f(\delta_n(x)) \neq g(\delta_n(x))$. Partitioning the disagreement set by fibers gives the equality. Surjectivity makes every fiber nonempty, proving the inequality, and constant fiber size gives the final formula. □

Remark 6.5 (Why no canonical refinement is implicit)**.** Definition 1.1 supplies no δ_n. A surjection exists between *finite nonempty* A_{n+1} and A_n because the former is larger when $|B| \geq 2$, but selecting one is extra structure. These choices are not automatically meaningful as physical subdivision maps, semantic abstractions, or verified knowledge updates. In a certified model, they must carry explicit interpretation and, where warranted, checking certificates.

Proposition 6.6 (A useful embedding that is not a refinement map)**.** *If $p \in P$, define $j_n : A_n \to A_{n+1}$ by*

$$j_n(a)(x) = \begin{cases} p, & x = a, \\ E, & x \neq a. \end{cases}$$

Then j_n is injective. Nevertheless j_n alone does not say that the level-$(n+1)$ representation preserves the meaning of a level-n configuration.

Proof. For distinct a, b, evaluating $j_n(a)$ and $j_n(b)$ at a distinguishes the two functions. The second claim is a modeling caution: no meaning-preserving relation was postulated. □

7 Relevance to Volume III.1 and certified knowledge

In Volume III.1, a complete occupancy report is distinguished from the hidden legal-rule state, a board transition, a history, and a verifier-accepted certificate. The new tower changes none of those distinctions. For each n, additional modeling data are needed to speak about lawfulness, observation, settlement, or epistemic gain.

Proposition 7.1 (No move and no knowledge theorem from the tower alone)**.** *There are two models with the same sets A_n and the same functions at every level but different legal-move relations and different verifier-accepted BANK contents. Therefore the recursion $A_{n+1} = [A_n \to B]$ entails no nontrivial transition, reachability, settlement, or monotone certified-knowledge assertion.*

Proof. Equip the common set-theoretic tower first with an empty transition relation and empty BANKs. Equip the same tower instead with any chosen nonempty transition relation and a fixed sound accepted record in a BANK, if a suitable certificate exists under the fixed theory. The recursive definitions are unchanged. Thus these extra properties cannot follow from the recursion alone. □

Research question. Can one give explicit maps δ_n, verifier-safe representations of their meaning, and charged algorithms that allow a hierarchy of reports to reduce settlement search cost relative to appropriate fixed-board baselines? The present paper defines the objects and proves their elementary structural properties; it does *not* claim such an acceleration theorem.

8 Conclusions, provenance, and next theorem target

The exact answer to "what kind of objects are A_n?" is: *iterated sets of total B-valued functions*, with level-n objects indexing the level-$(n+1)$ functions. With two labels they reproduce the iterated powerset construction up to explicit bijections; with several labels they resemble labeled partitions at each rank. Their cardinality increases strictly when at least two labels are available. But the original-square nested-occupant model and the board-of-reports model are distinct, and neither makes a claim about history or certified knowledge without extra structure. **Provenance and limits.**

The recurrence and the fixed E token originate in the motivating question of Brian Tenneson. The mathematical distinctions, proofs, countermodels, and alternative tower here were developed in this AI-assisted exposition. These are classical set-theoretic constructions, presented for the GBG+/Volume III.1 context; historical originality, computational advantage, and external peer review are not claimed.